Summarizing Distributions

X610 · Session 2 · *FoAS* Chapter 2

Today

- Expected values, moments, variance, and measures of dispersion
- Mean squared error and the expected value as the best constant predictor
- Covariance, correlation, conditional expectation, and conditional variance
- Conditional expectation functions, best linear predictors, and multivariate generalizations

Reading: *FoAS* Ch. 2; *Bernoulli's Fallacy* Ch. 2

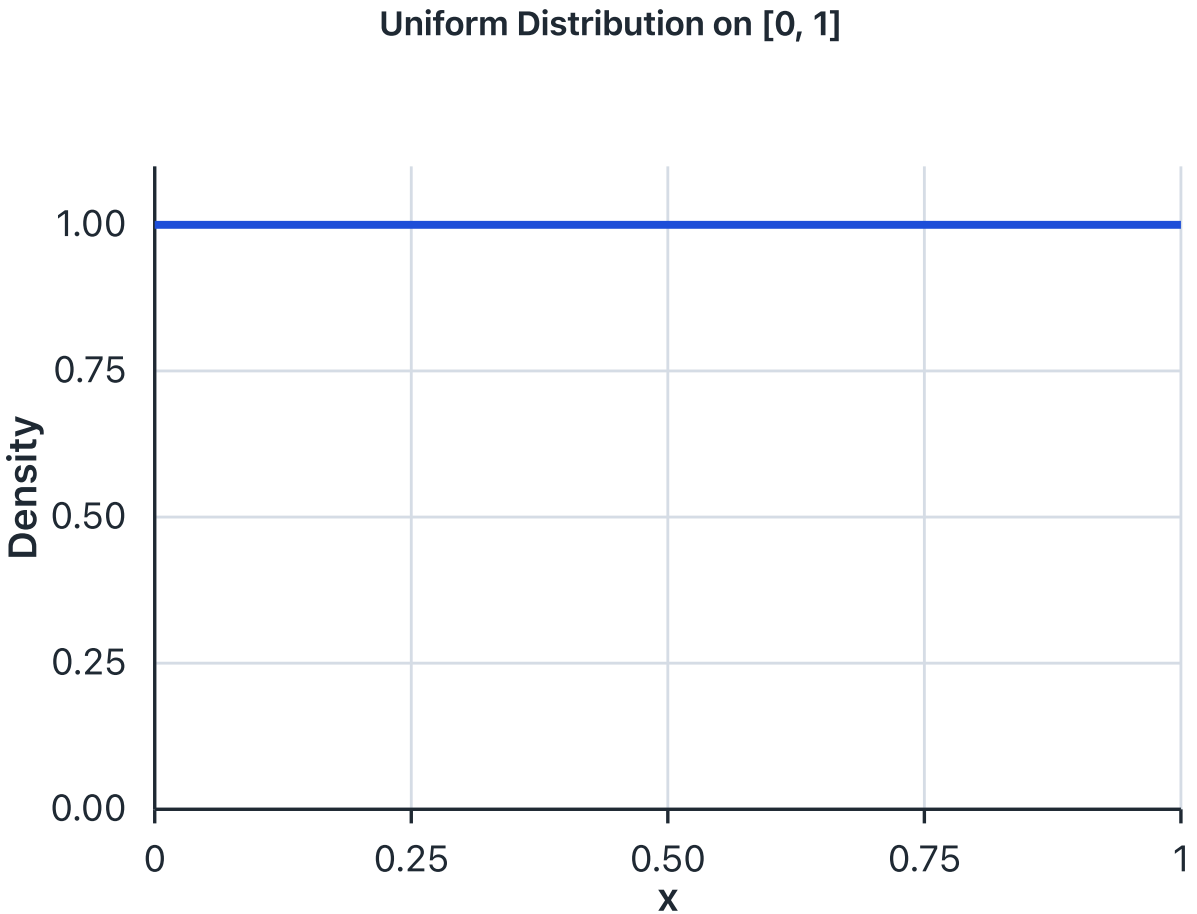
Expectation and moments

Notation: new this session

Notation	Definition and usage
$\arg\max$	The value, among all values in a particular set, that maximizes the given expression. For example, $\operatorname{argmax}_{x \in \mathbb{R}}(-x^2 - 2x + 3) = -1$. Assumes a unique maximum.
$\arg\min$	The value that minimizes the given expression. For example, $\operatorname{argmin}_{x \in \mathbb{R}}(x^2 - 4x + 1) = 2$. Assumes a unique minimum.

Three probability density functions

Distribution: Standard uniform



Expected value

Definition 2.1.1 — Expected Value

For a discrete random variable X with PMF f the expected value of X is

$$\mathrm{E}[X] = \sum_x x f(x).$$

For a continuous random variable X with PDF f, the expected value of X is

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx.$$

Regularity conditions in both cases ensure the sums/integrals converge absolutely.

Two simple examples

Example 2.1.2 — A Fair Die Roll

$$\mathrm{E}[X] = \sum_{x=1}^{6} x \cdot \frac{1}{6} = \frac{7}{2}.$$

Note that X never takes the value 3.5. A random variable need not take its expected value with positive probability.

Example 2.1.3 — Bernoulli Distribution

$$\mathrm{E}[X] = 0 \cdot (1-p) + 1 \cdot p = p,$$

so means of binary variables *are probabilities*.

Expectations of functions

Theorem 2.1.5 — Expectation of a Function of a Random Variable

If X is a discrete random variable with PMF f and g is a function of X, then

$$\mathrm{E}[g(X)] = \sum_x g(x) f(x).$$

If X is a continuous random variable with PDF f and g is a function of X, then

$$\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx.$$

- Just weight $g(x)$ by the original PMF/PDF
- In general $\mathrm{E}[g(X)] \neq g(\mathrm{E}[X])$

Linearity of expectations

Theorem 2.1.9 — Linearity of Expectations

Let X and Y be random variables. Then, $\forall a, b, c \in \mathbb{R}$,

$$\mathrm{E}[aX + bY + c] = a\mathrm{E}[X] + b\mathrm{E}[Y] + c.$$

- Holds with **no assumptions** about the relationship between X and Y — dependent, independent, … it doesn't matter

Linearity, verified

```
n = 1_000_000
X = rng.integers(1, 7, size=n)           # die roll
Y = X + rng.exponential(size=n)          # strongly dependent on X

print('E[2X + 3Y + 1] ≈', (2 * X + 3 * Y + 1).mean())
print('2E[X]+3E[Y]+1  ≈', 2 * X.mean() + 3 * Y.mean() + 1)
```

```
E[2X + 3Y + 1] ≈ 21.505361469279705
2E[X]+3E[Y]+1  ≈ 21.505361469279705
```

- Dependence between X and Y is irrelevant — linearity doesn't care

Moments

Definition 2.1.10 — j-th Raw Moment

For a random variable X and $j \in \mathbb{N}$, the j-th raw moment of X is $\mu'_j = \mathrm{E}[X^j]$.

Definition 2.1.11 — j-th Central Moment

For a random variable X and $j \in \mathbb{N}$, the j-th central moment of X is μ_j $= \mathrm{E}\left[(X - \mathrm{E}[X])^j\right]$.

- The first raw moment is the *expected value*;
  - the first central moment is always zero
- The second central moment has its own name…

Variance

Definition 2.1.12 — Variance

The variance of a random variable X is

$$\mathrm{V}[X] = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right].$$

Since $\mathrm{E}[X]$ is a constant, expand the square and apply linearity:

$$\begin{aligned}\mathrm{V}[X] &= \mathrm{E}\left[X^2 - 2X\mathrm{E}[X] + \mathrm{E}[X]^2\right] \\ &= \mathrm{E}[X^2] - 2\mathrm{E}[X]^2 + \mathrm{E}[X]^2 \\ &= \mathrm{E}[X^2] - \mathrm{E}[X]^2.\end{aligned}$$

Variance and standard deviation

Theorem 2.1.13 — Alternative Formula for Variance

For a random variable X,

$$\mathrm{V}[X] = \mathrm{E}[X^2] - \mathrm{E}[X]^2.$$

Definition 2.1.15 — Standard Deviation

$$\sigma[X] = \sqrt{\mathrm{V}[X]}$$

Die roll example

For a fair die, $\mathrm{E}[X] = 3.5$. Therefore,

$$\mathrm{V}[X] = \frac{1}{6}\sum_{x=1}^{6}(x-3.5)^2 = \tfrac{1}{6}\left(2\cdot 2.5^2 + 2\cdot 1.5^2 + 2\cdot 0.5^2\right)$$
$$= \tfrac{35}{12} \approx 2.92$$

Note also that

$$\mathrm{E}[X^2] = \tfrac{1}{6}(1+2^2+3^2+4^2+5^2+6^2) = \tfrac{91}{6},$$

so $\mathrm{V}[X] = \mathrm{E}[X^2] - \mathrm{E}[X]^2 = \tfrac{91}{6} - \tfrac{49}{4} = \tfrac{35}{12}.$

Die roll moments in python

```
X = stats.randint(1, 7)  # fair die: values 1 through 6
mean = X.moment(1)
second_moment = X.moment(2)

print('E[X] =', mean)
print('E[X²] =', second_moment)
print('V[X] from deviations =', X.expect(lambda x: (x - mean) ** 2))
print('V[X] from moments =', second_moment - mean ** 2)
```

```
E[X] = 3.5
E[X²] = 15.166666666666666
V[X] from deviations = 2.9166666666666665
V[X] from moments = 2.916666666666666
```

How variance and SD transform

Theorems 2.1.14 and 2.1.16 — Properties of Variance and Standard Deviation

For a random variable X, and $a, c \in \mathbb{R}$,

- $\mathrm{V}[X + c] = \mathrm{V}[X]$ and $\mathrm{V}[aX] = a^2\mathrm{V}[X]$
- $\sigma[X + c] = \sigma[X]$ and $\sigma[aX] = |a|\,\sigma[X]$

- a *shift* doesn't change the *spread* of a random variable
- but *scaling* does:
  - scales SD linearly, but variance increases quadratically

Exercise

Let $X \sim B(p)$. What is the variance of X?

Chebyshev's inequality

Theorem 2.1.18 — Chebyshev's Inequality

Let X be a random variable with finite $\sigma[X] > 0$. Then, $\forall \epsilon > 0$,

$$\Pr\left[|X - \mathrm{E}[X]| \geq \epsilon\, \sigma[X]\right] \leq \frac{1}{\epsilon^2}.$$

- Knowing only the mean and SD bounds the tails for **any** distribution
- At most $\frac{1}{4}$ of probability lies beyond 2 SDs; at most $\frac{1}{9}$ beyond 3 SDs

How tight is Chebyshev?

```
tight = stats.rv_discrete(values=([-2, 0, 2], [1/8, 3/4, 1/8]))
dists = {'normal': stats.norm(), 'uniform': stats.uniform(),
         'exponential': stats.expon(), 'three-point': tight}

print('eps=2: bound = 0.25')
for name, d in dists.items():
    m, s = d.mean(), d.std()
    actual = d.expect(lambda x: np.abs(x - m) >= 2 * s)
    print(f'  {name:12s} actual = {actual:.4f}')
```

```
eps=2: bound = 0.25
  normal       actual = 0.0455
  uniform      actual = 0.0000
  exponential  actual = 0.0498
  three-point  actual = 0.2500
```

- Familiar distributions fall well below the bound
- The specially constructed 3-point distribution attains it exactly

The normal distribution

Definition 2.1.19 — Normal Distribution

A continuous random variable X follows a normal distribution if it has PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \ \forall x \in \mathbb{R},$$

for some constants $\mu, \sigma \in \mathbb{R}$ with $\sigma > 0$. We write $X \sim N(\mu, \sigma^2)$.

Theorem 2.1.20 — Mean and Standard Deviation of the Normal Distribution

If $X \sim N(\mu, \sigma^2)$, then $\mathrm{E}[X] = \mu$ and $\sigma[X] = \sigma$.

- Two parameters characterize the distribution:
  - μ shifts it and σ scales it

Normals stay normal

Theorem 2.1.21 — Properties of the Normal Distribution

Suppose $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. Then

- $\forall a, b \in \mathbb{R}$ with $a \neq 0$, if $W = aX + b$, then $W \sim N(a\mu_X + b, a^2\sigma_X^2)$.
- If $X \perp\!\!\!\perp Y$ and $Z = X + Y$, then $Z \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

- Linear combinations of independent normals are normal
- Standardizing: $\frac{X-\mu}{\sigma} \sim N(0, 1)$

A question

You must guess the value of a random variable X before it is drawn, and you'll be penalized by the *squared* error of your guess.

What single number should you guess?

- The most likely value? The middle value? Something else?
- Does your answer depend on the distribution's shape?

Mean squared error

Definition 2.1.22 — Mean Squared Error (MSE) about c

For a random variable X and $c \in \mathbb{R}$, the mean squared error of X about c is $\mathrm{E}\left[(X-c)^2\right]$.

Theorem 2.1.23 — Alternative Formula for MSE

For a random variable X and $c \in \mathbb{R}$,

$$\mathrm{E}\left[(X-c)^2\right] = \mathrm{V}[X] + (\mathrm{E}[X]-c)^2.$$

- MSE = irreducible variance + squared *bias* of our guess

The answer: the expected value

Theorem 2.1.24 — The Expected Value Minimizes MSE

The value of c that minimizes the MSE of r.v. X about c is $c = \mathrm{E}[X]$.

```
X = rng.exponential(size=200_000)        # skewed: E[X]=1, median≈0.69

cs = np.linspace(0.2, 2.0, 100)
mse = [((X - c) ** 2).mean() for c in cs]
print('argmin MSE ≈', cs[np.argmin(mse)], '   E[X] ≈', X.mean())
```

```
argmin MSE ≈ 1.0    E[X] ≈ 0.9998875768385554
```

- Even for a skewed distribution, the mean—not the median or mode—is best under squared error
- Exercise: what if we minimize mean *absolute* deviation instead?

Expectation and moments recap

- Expectation: probability-weighted average; linear regardless of dependence
- Variance/SD measure spread; SD preserves the scale of the data
- Chebyshev: mean + SD bound the tails of any distribution
- The normal is closed under linear combinations; (μ, σ) are all we need to know
- $\mathrm{E}[X]$ is the best constant predictor under MSE

Conditional expectation and best prediction

Conditional expectation and best prediction: topics

- Covariance and correlation
- Independence vs. zero correlation
- Conditional expectation and the CEF
- The Law of Iterated Expectations; Law of Total Variance
- The CEF as best predictor; the best linear predictor (BLP)

Reading: *FoAS* Ch. 2 (§2.2–2.3)

Covariance

Definition 2.2.1 — Covariance

The covariance of two random variables X and Y is

$$\operatorname{Cov}[X, Y] = \mathrm{E}\big[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])\big].$$

Theorem 2.2.2 — Alternative Formula for Covariance

For random variables X and Y,

$$\operatorname{Cov}[X, Y] = \mathrm{E}[XY] - \mathrm{E}[X]\mathrm{E}[Y].$$

The variance rule

Theorem 2.2.3 — Variance Rule

Let X and Y be random variables. Then

$$\mathrm{V}[X+Y] = \mathrm{V}[X] + 2\,\mathrm{Cov}[X,Y] + \mathrm{V}[Y].$$

More generally, $\forall a, b, c \in \mathbb{R}$,

$$\mathrm{V}[aX+bY+c] = a^2\mathrm{V}[X] + 2ab\,\mathrm{Cov}[X,Y] + b^2\mathrm{V}[Y].$$

- Variances are *not* linear — the cross term drives diversification, clustering, and much else
- Mnemonic: $(x+y)^2 = x^2 + 2xy + y^2$

Correlation

Definition 2.2.5 — Correlation

The correlation of two random variables X and Y with $\sigma[X] > 0$ and $\sigma[Y] > 0$ is

$$\rho[X, Y] = \frac{\mathrm{Cov}[X, Y]}{\sigma[X]\sigma[Y]}.$$

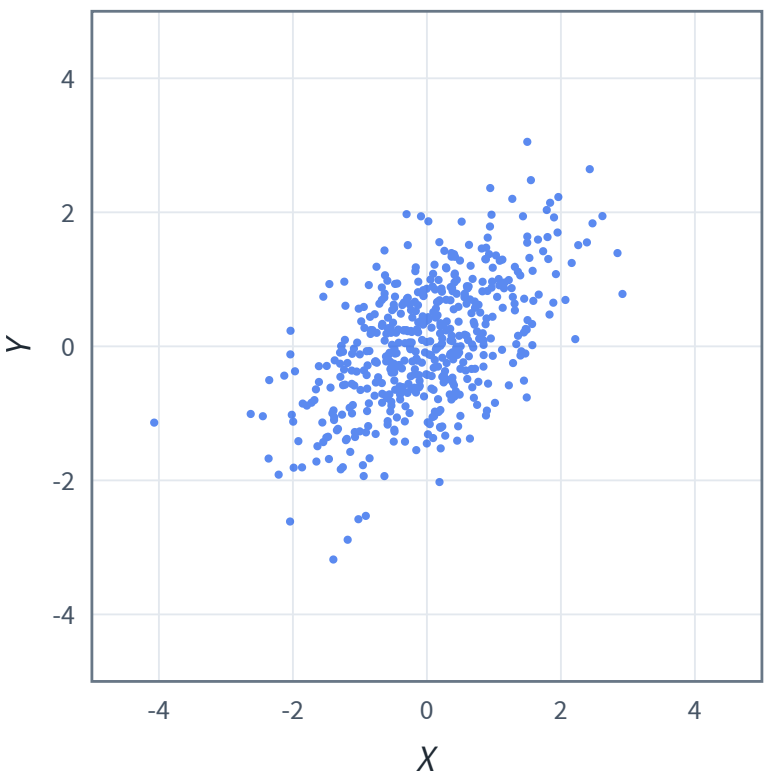
Correlation is scale-free.

Correlated data

Correlation ρ 0.60

SD(Y) / SD(X) 1.0

Draw again



500 draws; ρ is the population value. At ±1, vary SD(Y) / SD(X).

Correlation and linear dependence

Theorem 2.2.6 — Correlation and Linear Dependence

For random variables X and Y,

- $\rho[X, Y] \in [-1, 1]$.
- $\rho[X, Y] = 1 \iff \exists\, a, b \in \mathbb{R}$ with $b > 0$ such that $Y = a + bX$.
- $\rho[X, Y] = -1 \iff \exists\, a, b \in \mathbb{R}$ with $b > 0$ such that $Y = a - bX$.

Independence kills covariance

Theorem 2.2.8 — Implications of Independence (Part II)

If X and Y are independent random variables, then

- $\mathrm{E}[XY] = \mathrm{E}[X]\mathrm{E}[Y]$.
- Covariance is zero: $\mathrm{Cov}[X, Y] = 0$.
- Correlation is zero: $\rho[X, Y] = 0$.
- Variances are additive: $\mathrm{V}[X + Y] = \mathrm{V}[X] + \mathrm{V}[Y]$.

A question

We just saw: independence $\Rightarrow$ zero correlation.

Does zero correlation imply independence?

- If yes: why?
- If no: construct a counterexample — two variables with $\rho = 0$ that are clearly related

No: zero correlation ≠ independence

Example 2.2.9 — Zero Correlation Does Not Imply Independence

Let X take values $\{-1, 0, 1\}$ with probability $\frac{1}{3}$ each, and let $Y = X^2$. Then Y is a *deterministic function* of X, yet $\mathrm{Cov}[X, Y] = \mathrm{E}[XY] - \mathrm{E}[X]\mathrm{E}[Y] = 0 - 0 = 0$.

```
X = rng.integers(-1, 2, size=1_000_000)
Y = X ** 2

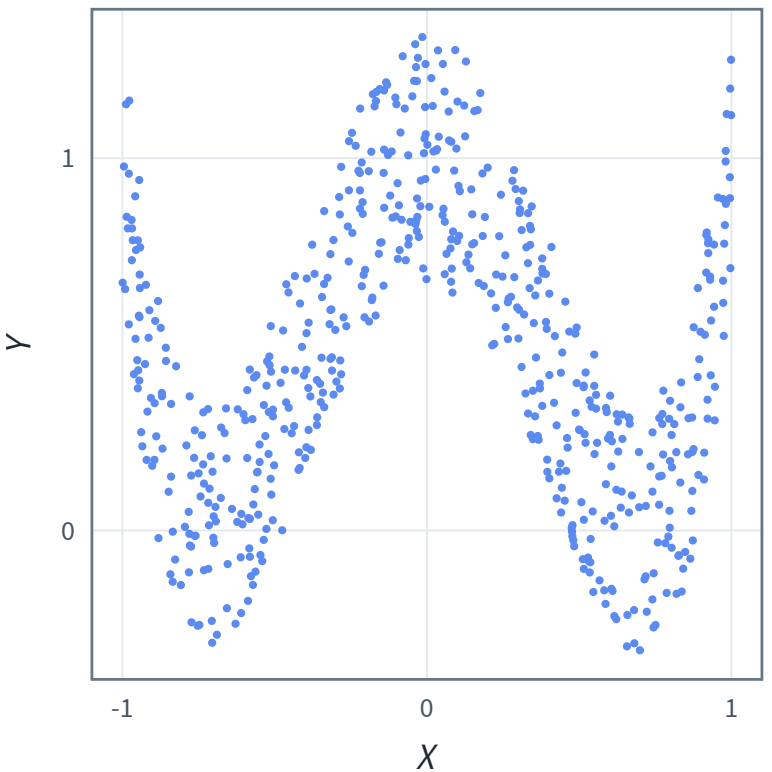
print('corr(X, Y)     ≈', np.corrcoef(X, Y)[0, 1].round(4))
print('E[Y | X = 0]   =', Y[X == 0].mean(), '  E[Y | X = ±1] =', Y[X != 0].mean
```

```
corr(X, Y)     ≈ 0.0001
E[Y | X = 0]   = 0.0   E[Y | X = ±1] = 1.0
```

- Correlation sees only *linear* structure; the dependence here is perfectly symmetric, so it cancels

Zero correlation can hide patterns

Pattern: W Draw again



Population ρ = 0; sample r = -0.01.

But also: causation ⇏ correlation



Post

Rafe Meager (they/them)
@economeager

Ok so chapter 12 of “Noise” by Kahneman, Sibony and Sunstein contains a repeated, incorrect claim about stats / causality: the claim that no correlation implies no causation. Easy to disprove this claim, no math needed, see my next tweet.

lation.
We must, however, remember that while correlation does not imply causation, causation does imply correlation. Where there is a causal link, we should find a correlation. If you find no correlation between age and shoe size among adults, then you can safely conclude that after the end of adolescence, age does not make feet grow larger and that you have to look elsewhere for the causes of differences in shoe size.
In short, wherever there is causality, there is correlation. It follows that where there is causality, we should be able to predict—and correlation, the accuracy of this prediction, is a measure of how much causality we understand. Hence the conclusion of the Princeton researchers is this: the extent to which sociologists can predict events like evictions, as measured by a correlation of .22, is an indication of how much—or how little—they understand about the life trajectories of these families. Objective ignorance sets a ceiling not only on our predic-

on does not imply c
sation does imply

1:19 PM · May 21, 2021

87 503 1.8K 380

Rafe Meager (they/them) @economeager · May 21, 2021
Imagine driving a car, reaching a hill and pumping the gas as you begin to go up so that your speed is constant. The correlation between pressing on the gas and the speed of the car is zero but they’re obviously causally related, it’s that the agent is optimizing speed!

66 69 908

→ More discussion on this blog

Conditional expectation

Definition 2.2.10 — Conditional Expectation

For discrete random variables X and Y with joint PMF f, the conditional expectation of Y given $X = x$ is

$$\mathrm{E}[Y \mid X = x] = \sum_{y} y f_{Y|X}(y \mid x), \ \forall x \in \mathrm{Supp}[X].$$

For jointly continuous random variables X and Y with joint PDF f,

$$\mathrm{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x)\, dy, \ \forall x \in \mathrm{Supp}[X].$$

The conditional expectation function

Definition 2.2.15 — Conditional Expectation Function (CEF)

For random variables X and Y with joint PMF/PDF f, the conditional expectation function of Y given $X = x$ is

$$G_Y(x) = \mathrm{E}[Y \mid X = x], \ \forall x \in \mathrm{Supp}[X].$$

- One function collecting every conditional expectation
- $\mathrm{E}[Y \mid X]$ denotes $G_Y(X)$: a *random variable* whose value is determined by X

The Law of Iterated Expectations

Theorem 2.2.17 — Law of Iterated Expectations

For random variables X and Y,

$$\mathrm{E}[Y] = \mathrm{E}\big[\mathrm{E}[Y \mid X]\big].$$

- $\mathrm{E}[Y \mid X]$ answers "what would I expect Y to be if I knew X?" Since we don't yet know X, this answer is itself a random variable.
- The LIE says that averaging these conditional forecasts over all possible values of X gives the unconditional forecast: $\mathrm{E}\big[\mathrm{E}[Y \mid X]\big] = \mathrm{E}[Y]$.
- Equivalently: my best guess today of what I'll forecast once I learn X is just my forecast today.

Law of Iterated Expectations example

```
n = 1_000_000
X = (rng.random(n) < 0.5).astype(int)                    # coin
Y = np.where(X == 0, rng.integers(1, 5, n), rng.integers(1, 7, n))

print('E[Y]                ≈', Y.mean())
print('E[E[Y|X]] = ½·5/2 + ½·7/2 =', 0.5 * 2.5 + 0.5 * 3.5)
```

```
E[Y]                ≈ 2.999794
E[E[Y|X]] = ½·5/2 + ½·7/2 = 3.0
```

The Law of Total Variance

- The law of total variance (Thm 2.2.18) splits total uncertainty about Y into two pieces:

$$\operatorname{Var}(Y) = \underbrace{E\big[\operatorname{Var}(Y \mid X)\big]}_{\text{uncertainty left after learning } X} + \underbrace{\operatorname{Var}\Big(E[Y \mid X]\Big)}_{\text{uncertainty about what my forecast will be}}$$

  - First term: Noise that knowing X can't remove, averaged over possible values of X
  - Second term: How much my forecast of Y moves around as X changes. This is the part of the variation in Y that X explains.
- Learning X can only *remove* uncertainty: $E\big[\operatorname{Var}(Y \mid X)\big] \leq \operatorname{Var}(Y)$.
- $\operatorname{Var}(Y \mid X)$ answers "how uncertain would I still be about Y if I knew X?"
  - Like $E[Y \mid X]$, this is a random variable, because we don't yet know X.

Deviations from the CEF

Theorem 2.2.19 — Properties of Deviations from the CEF

Let X and Y be random variables and let $\varepsilon = Y - \mathrm{E}[Y \mid X]$. Then

- $\mathrm{E}[\varepsilon \mid X] = 0$.
- $\mathrm{E}[\varepsilon] = 0$.
- If g is a function of X, then $\mathrm{Cov}[g(X), \varepsilon] = 0$.
- $\mathrm{V}[\varepsilon \mid X] = \mathrm{V}[Y \mid X]$.
- $\mathrm{V}[\varepsilon] = \mathrm{E}[\mathrm{V}[Y \mid X]]$.

- What's left after subtracting the CEF is pure noise with respect to X — uncorrelated with *any* function of it

The CEF is the best predictor

Theorem 2.2.20 — The CEF is the Best Predictor

For random variables X and Y, the CEF, $\mathrm{E}[Y \mid X]$, is the best (minimum MSE) predictor of Y given X.

- Earlier today: the best *constant* guess is $\mathrm{E}[Y]$
- Now: best *function of* X is $\mathrm{E}[Y \mid X]$ — no assumptions
- But: without assumptions, the CEF can be quite complicated
  - Later you'll see that *linear regression* imposes a linear structure on the CEF, which is often a good approximation

The best linear predictor

Theorem 2.2.21 — Best Linear Predictor (BLP)

For random variables X and Y, if $\mathrm{V}[X] > 0$, then the best (minimum MSE) linear predictor of Y given X is $g(X) = \alpha + \beta X$, where

$$\alpha = \mathrm{E}[Y] - \frac{\mathrm{Cov}[X, Y]}{\mathrm{V}[X]}\mathrm{E}[X], \qquad \beta = \frac{\mathrm{Cov}[X, Y]}{\mathrm{V}[X]}.$$

Theorem 2.2.22 — Properties of Deviations from the BLP

Let $\varepsilon = Y - g(X)$, where $g(X)$ is the BLP. Then $\mathrm{E}[\varepsilon] = 0$, $\mathrm{E}[X\varepsilon] = 0$, and $\mathrm{Cov}[X, \varepsilon] = 0$.

CEF vs. BLP

```
n = 2_000
X = rng.random(n)                          # U(0,1)
Y = 10 * X ** 2 + rng.standard_normal(n)

b = np.cov(X, Y)[0, 1] / X.var()           # sample analogs of beta, alpha
a = Y.mean() - b * X.mean()

xs = np.linspace(0, 1, 100)
fig, ax = plt.subplots(figsize=(7, 3.4))
ax.plot(X, Y, '.', ms=2, alpha=0.3)
ax.plot(xs, 10 * xs ** 2, 'k', lw=2, label=r'CEF: $10x^2$')
ax.plot(xs, a + b * xs, 'r', lw=2, label=f'BLP: {a:.2f} + {b:.2f}x')
ax.set(xlabel='x', ylabel='y')
ax.legend()
plt.show()
```

The BLP depends on the dist'n of X

- With a *nonlinear* CEF, changing the distribution of X changes the BLP (Example 2.2.24)
- The BLP is a property of the joint distribution, not of the CEF alone
- Practical warning: a "slope" estimated where data are dense may extrapolate badly where data are sparse

Independence, again

Theorem 2.2.25 — Implications of Independence (Part III)

If X and Y are independent random variables, then

- $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
- $\mathrm{V}[Y \mid X] = \mathrm{V}[Y]$.
- The BLP of Y given X is $\mathrm{E}[Y]$.

- Under independence, conditioning is useless: the best prediction ignores X entirely

Covariance matrices

Definition 2.3.1 — Covariance Matrix

For a random vector $\mathbf{X}$ of length K, the covariance matrix $\mathrm{V}[\mathbf{X}]$ is the matrix whose (k, k')-th entry is $\mathrm{Cov}[X_{[k]}, X_{[k']}]$.

$$\mathrm{V}[\mathbf{X}] = \begin{pmatrix} \mathrm{V}[X_1] & \mathrm{Cov}[X_1, X_2] & \cdots & \mathrm{Cov}[X_1, X_K] \\ \mathrm{Cov}[X_2, X_1] & \mathrm{V}[X_2] & \cdots & \mathrm{Cov}[X_2, X_K] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}[X_K, X_1] & \mathrm{Cov}[X_K, X_2] & \cdots & \mathrm{V}[X_k] \end{pmatrix}$$

- The covariance matrix is symmetric; the diagonal entries are the variances of the individual components.